

图像配准实验平台测试用例报告

本报告优先根据 outputs/batch 下各 pipeline 的 summary.csv 自动整理；不同方法族使用各自的 CSV schema，旧输出缺少 summary.csv 时才回退读取 run_summary.json。

1. 总体统计

指标	数值
测试用例总数	96
通过用例	53
未通过用例	43
Pipeline 数量	8
方法族数量	4
样本数量	12
平均耗时	900.9 ms

2. 批量测试入口

测试类型	配置文件	运行命令
点特征法批量测试	configs/pipeline/batch/batch_keypoint.yaml	build-mingw\registration_app.exe configs\pipeline\batch\batch_keypoint.yaml
结构特征法批量测试	configs/pipeline/batch/batch_structure.yaml	build-mingw\registration_app.exe configs\pipeline\batch\batch_structure.yaml
直接法批量测试	configs/pipeline/batch/batch_direct.yaml	build-mingw\registration_app.exe configs\pipeline\batch\batch_direct.yaml
深度学习匹配批量测试	configs/pipeline/batch/batch_learning.yaml	build-mingw\registration_app.exe configs\pipeline\batch\batch_learning.yaml

3. 数据集样本

样本	源图	目标图	源图大小	目标图大小	已执行用例数
Test01	datasets/Test01/source.png	datasets/Test01/target.png	24.4 KB	97.1 KB	8
Test02	datasets/Test02/moving.png	datasets/Test02/reference.png	24.5 KB	77.5 KB	8
Test03	datasets/Test03/source.png	datasets/Test03/target.png	6.8 KB	19.8 KB	8
Test04	datasets/Test04/source.png	datasets/Test04/target.png	4.1 KB	18.9 KB	8
Test05	datasets/Test05/source.png	datasets/Test05/target.png	4.6 KB	7.0 KB	8
Test06	datasets/Test06/source.png	datasets/Test06/target.png	5.1 KB	14.0 KB	8
Test07	datasets/Test07/source.png	datasets/Test07/target.png	45.6 KB	49.6 KB	8
Test08	datasets/Test08/source.png	datasets/Test08/target.png	21.8 KB	25.0 KB	8
Test09	datasets/Test09/source.png	datasets/Test09/target.png	22.1 KB	28.8 KB	8
Test10	datasets/Test10/source.png	datasets/Test10/target.png	14.1 KB	12.0 KB	8
Test11	datasets/Test11/source.png	datasets/Test11/target.png	17.1 KB	28.0 KB	8
Test12	datasets/Test12/source.png	datasets/Test12/target.png	12.9 KB	18.1 KB	8

4. Pipeline 汇总

方法族	Pipeline	用例数	通过数	平均 IoU	平均内点/置信度	平均 SSIM	平均耗时 ms	CSV 汇总
direct	global_direct_pipeline	12	8 / 12	0.694	0.657	0.847	254.4	outputs/batch/direct/global_direct_pipeline/summary.csv
keypoint	sift_pipeline	12	7 / 12	0.623	0.230	0.794	125.3	outputs/batch/keypoint/sift_pipeline/summary.csv
learning	lofr_learning_pipeline	12	10 / 12	0.674	0.450	0.755	1889.5	outputs/batch/learning/lofr_learning_pipeline/summary.csv
learning	superpoint_lightglue_learning_pipeline	12	5 / 12	0.705	0.458	0.840	3584.0	outputs/batch/learning/superpoint_lightglue_learning_pipeline/summary.csv

learning	superpoint_superglue_learning_pipeline	12	10 / 12	0.626	0.538	0.788	1169.7	outputs/batch/learning/superpoint_superglue_learning_pipeline/summary.csv
structure	contour_pipeline	12	2 / 12	0.779	0.167	0.828	11.2	outputs/batch/structure/contour_pipeline/summary.csv
structure	edge_pipeline	12	2 / 12	0.890	0.398	0.794	113.2	outputs/batch/structure/edge_pipeline/summary.csv
structure	line_pipeline	12	9 / 12	0.559	0.592	0.703	60.2	outputs/batch/structure/line_pipeline/summary.csv

5. 测试用例明细

样本	方法族	Pipeline	状态	说明	对象数量	候选/原始匹配	过滤匹配	内点	内点率/置信度	重投影误差	IoU	SSIM
Test01	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.267	0.000	0.568	0.687
Test02	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.487	0.000	0.525	0.775
Test03	direct	global_direct_pipeline	未通过	warp overlap IoU below threshold: 0.044046 < 0.200000	对应点数0	0	0	1	0.353	0.000	0.044	-
Test04	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.865	0.000	0.720	0.924
Test05	direct	global_direct_pipeline	未通过	estimation failed: findTransformECC failed: OpenCV(4.12.0) D:\Opcv\opencv\sources\modules\video\src\ecc.cpp:589: error: (-7:Iterations do not converge) The algorithm stopped before its convergence. The correlation is going to be minimized. Images may be uncorrelated or non-overlapped in function 'findTransformECC'	对应点数0	0	0	0	0.000	0.000	-	-
Test06	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.985	0.000	0.975	0.937
Test07	direct	global_direct_pipeline	未通过	warp photometric error above threshold: 0.171296 > 0.150000	对应点数0	0	0	1	0.676	0.000	0.645	-
Test08	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.982	0.000	0.876	0.892
Test09	direct	global_direct_pipeline	未通过	warp photometric error above threshold: 0.175144 > 0.150000	对应点数0	0	0	1	0.650	0.000	0.669	-
Test10	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.780	0.000	0.826	0.767
Test11	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.872	0.000	0.818	0.876
Test12	direct	global_direct_pipeline	通过	OK	对应点数0	0	0	1	0.963	0.000	0.965	0.914
Test01	keypoint	sift_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	关键点数97 / 133	24	24	0	0.000	0.000	-	-
Test02	keypoint	sift_pipeline	通过	OK	关键点数115 / 158	40	40	5	0.125	0.000	0.476	0.804
Test03	keypoint	sift_pipeline	通过	OK	关键点数	27	27	9	0.333	0.000	0.127	0.952

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Test04	keypoint	sift_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	关键点数 38 / 166	20	20	0	0.000	0.000	-	-	
Test05	keypoint	sift_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	关键点数 35 / 36	12	12	0	0.000	0.000	-	-	
Test06	keypoint	sift_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	关键点数 51 / 69	20	20	0	0.000	0.000	-	-	
Test07	keypoint	sift_pipeline	通过	OK	关键点数 353 / 244	108	108	68	0.630	0.000	0.904	0.934	
Test08	keypoint	sift_pipeline	通过	OK	关键点数 191 / 159	66	66	42	0.636	0.000	0.876	0.891	
Test09	keypoint	sift_pipeline	通过	OK	关键点数 301 / 156	52	52	14	0.269	0.000	0.681	0.764	
Test10	keypoint	sift_pipeline	通过	OK	关键点数 57 / 54	13	13	5	0.385	0.000	0.879	0.869	
Test11	keypoint	sift_pipeline	通过	OK	关键点数 211 / 106	56	56	11	0.196	0.000	0.760	0.818	
Test12	keypoint	sift_pipeline	未通过	metric quality validation failed: SSIM 0.323344 < 0.600000	关键点数 115 / 74	27	27	5	0.185	0.000	0.285	0.323	
Test01	learning	loftr_learning_pipeline	通过	OK	学习点数 211 / 211	211	211	45	0.213	0.000	0.493	0.661	
Test02	learning	loftr_learning_pipeline	通过	OK	学习点数 257 / 257	257	257	87	0.339	0.000	0.489	0.749	
Test03	learning	loftr_learning_pipeline	通过	OK	学习点数 49 / 49	49	49	16	0.327	0.000	0.399	0.615	
Test04	learning	loftr_learning_pipeline	通过	OK	学习点数 94 / 94	94	94	51	0.543	0.000	0.696	0.898	
Test05	learning	loftr_learning_pipeline	通过	OK	学习点数 29 / 29	29	29	3	0.103	0.000	0.314	0.475	
Test06	learning	loftr_learning_pipeline	通过	OK	学习点数 84 / 84	84	84	71	0.845	0.000	0.969	0.906	
Test07	learning	loftr_learning_pipeline	未通过	warp photometric error above threshold: 0.226008 > 0.150000	学习点数 221 / 221	221	221	26	0.118	0.000	0.946	-	
Test08	learning	loftr_learning_pipeline	通过	OK	学习点数 921 / 921	921	921	911	0.989	0.000	0.923	0.811	

Test09	learning	loftr_learning_pipeline	未通过	warp photometric error above threshold: 0.174039 > 0.150000	学习 点数 228 / 228	228	228	54	0.237	0.000	0.457	-
Test10	learning	loftr_learning_pipeline	通过	OK	学习 点数 52 / 52	52	52	17	0.327	0.000	0.646	0.649
Test11	learning	loftr_learning_pipeline	通过	OK	学习 点数 90 / 90	90	90	35	0.389	0.000	0.813	0.874
Test12	learning	loftr_learning_pipeline	通过	OK	学习 点数 859 / 859	859	859	833	0.970	0.000	0.949	0.909
Test01	learning	superpoint_lightglue_learning_pipeline	通过	OK	学习 点数 146 / 146	146	146	53	0.363	0.000	0.376	0.587
Test02	learning	superpoint_lightglue_learning_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	学习 点数 6 / 6	6	6	0	0.000	0.000	-	-
Test03	learning	superpoint_lightglue_learning_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	学习 点数 6 / 6	6	6	0	0.000	0.000	-	-
Test04	learning	superpoint_lightglue_learning_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	学习 点数 36 / 36	36	36	0	0.000	0.000	-	-
Test05	learning	superpoint_lightglue_learning_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	学习 点数 14 / 14	14	14	0	0.000	0.000	-	-
Test06	learning	superpoint_lightglue_learning_pipeline	通过	OK	学习 点数 79 / 79	79	79	79	1.000	0.000	0.972	0.918
Test07	learning	superpoint_lightglue_learning_pipeline	未通过	warp photometric error above threshold: 0.157681 > 0.150000	学习 点数 83 / 83	83	83	15	0.181	0.000	0.405	-
Test08	learning	superpoint_lightglue_learning_pipeline	通过	OK	学习 点数 430 / 430	430	430	430	1.000	0.000	0.934	0.892
Test09	learning	superpoint_lightglue_learning_pipeline	未通过	warp photometric error above threshold: 0.186451 > 0.150000	学习 点数 3 / 3	3	3	3	1.000	0.000	0.481	-
Test10	learning	superpoint_lightglue_learning_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	学习 点数 11 / 11	11	11	0	0.000	0.000	-	-
Test11	learning	superpoint_lightglue_learning_pipeline	通过	OK	学习 点数 147 / 147	147	147	141	0.959	0.000	0.818	0.891
Test12	learning	superpoint_lightglue_learning_pipeline	通过	OK	学习 点数 411 / 411	411	411	410	0.998	0.000	0.951	0.913
Test01	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 70 / 70	70	70	15	0.214	0.000	0.375	0.584
Test02	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 53 / 53	53	53	18	0.340	0.000	0.555	0.810

Test03	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 8 / 8	8	8	3	0.375	0.000	0.320	0.641
Test04	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 25 / 25	25	25	6	0.240	0.000	0.668	0.904
Test05	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 3 / 3	3	3	2	0.667	0.000	0.458	0.563
Test06	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 64 / 64	64	64	58	0.906	0.000	0.975	0.934
Test07	learning	superpoint_superglue_learning_pipeline	未通过	warp overlap IoU below threshold: 0.018238 < 0.200000	学习 点数 34 / 34	34	34	13	0.382	0.000	0.018	-
Test08	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 154 / 154	154	154	154	1.000	0.000	0.933	0.887
Test09	learning	superpoint_superglue_learning_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	学习 点数 11 / 11	11	11	0	0.000	0.000	-	-
Test10	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 8 / 8	8	8	4	0.500	0.000	0.820	0.756
Test11	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 65 / 65	65	65	55	0.846	0.000	0.817	0.887
Test12	learning	superpoint_superglue_learning_pipeline	通过	OK	学习 点数 94 / 94	94	94	93	0.989	0.000	0.951	0.914
Test01	structure	contour_pipeline	未通过	associate failed: not enough contour matches: 1	结构 数 21 / 14	0	0	0	0.000	0.000	-	-
Test02	structure	contour_pipeline	未通过	associate failed: not enough contour matches: 1	结构 数 23 / 25	0	0	0	0.000	0.000	-	-
Test03	structure	contour_pipeline	未通过	associate failed: not enough contour matches: 0	结构 数 1 / 4	0	0	0	0.000	0.000	-	-
Test04	structure	contour_pipeline	未通过	associate failed: not enough contour matches: 1	结构 数 6 / 55	0	0	0	0.000	0.000	-	-
Test05	structure	contour_pipeline	未通过	associate failed: not enough contour matches: 1	结构 数 6 / 7	0	0	0	0.000	0.000	-	-
Test06	structure	contour_pipeline	未通过	associate failed: not enough contour matches: 1	结构 数 3 / 2	0	0	0	0.000	0.000	-	-
Test07	structure	contour_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	结构 数 43 / 55	43	2	0	0.000	0.000	-	-
Test08	structure	contour_pipeline	通过	OK	结构 数 35 / 32	35	7	7	1.000	0.000	0.876	0.890
Test09	structure	contour_pipeline	通过	OK	结构 数 23 / 18	23	4	4	1.000	0.000	0.681	0.766

Test10	structure	contour_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	结构数 14 / 15	14	2	0	0.000	0.000	-	-
Test11	structure	contour_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	结构数 10 / 9	10	2	0	0.000	0.000	-	-
Test12	structure	contour_pipeline	未通过	estimation failed: failed to refine rigid transform from OpenCV RANSAC inliers	结构数 14 / 19	14	3	0	0.000	0.000	-	-
Test01	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 332.340637	结构数 15071 / 16900	0	0	0	0.000	0.000	-	-
Test02	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 218.695419	结构数 12997 / 19503	0	0	0	0.000	0.000	-	-
Test03	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 38.845886	结构数 5446 / 5493	0	0	0	0.000	0.000	-	-
Test04	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 8.595016	结构数 11894 / 12194	0	0	0	0.000	0.000	-	-
Test05	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 42.975082	结构数 12259 / 5769	0	0	0	0.000	0.000	-	-
Test06	structure	edge_pipeline	通过	OK	结构数 10621 / 10852	1	1	10621	0.955	0.000	0.949	0.854
Test07	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 69.715134	结构数 20660 / 19325	0	0	0	0.000	0.000	-	-
Test08	structure	edge_pipeline	通过	OK	结构数 11477 / 10246	1	1	11477	3.820	0.000	0.831	0.734
Test09	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 42.722244	结构数 16214 / 11864	0	0	0	0.000	0.000	-	-
Test10	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 24.830048	结构数 11947 / 8713	0	0	0	0.000	0.000	-	-
Test11	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 50.615097	结构数 17366 / 11622	0	0	0	0.000	0.000	-	-
Test12	structure	edge_pipeline	未通过	associate failed: hausdorff score above threshold: 52.665237	结构数 5468 / 6178	0	0	0	0.000	0.000	-	-

Test01	structure	line_pipeline	通过	OK	结构数 162 / 242	0	62	57	0.919	0.000	0.461	0.639
Test02	structure	line_pipeline	通过	OK	结构数 77 / 102	0	39	35	0.897	0.000	0.522	0.769
Test03	structure	line_pipeline	未通过	warp overlap IoU below threshold: 0.038033 < 0.200000	结构数 37 / 27	0	10	6	0.600	0.000	0.038	-
Test04	structure	line_pipeline	通过	OK	结构数 65 / 118	0	53	47	0.887	0.000	0.686	0.892
Test05	structure	line_pipeline	通过	OK	结构数 62 / 23	0	13	8	0.615	0.000	0.214	0.550
Test06	structure	line_pipeline	通过	OK	结构数 47 / 52	0	32	29	0.906	0.000	0.925	0.751
Test07	structure	line_pipeline	通过	OK	结构数 207 / 191	0	49	17	0.347	0.000	0.468	0.720
Test08	structure	line_pipeline	通过	OK	结构数 69 / 51	0	30	19	0.633	0.000	0.756	0.596
Test09	structure	line_pipeline	通过	OK	结构数 77 / 73	0	4	2	0.500	0.000	0.586	0.601
Test10	structure	line_pipeline	未通过	associate failed: not enough candidate line matches: 0	结构数 59 / 56	0	0	0	0.000	0.000	-	-
Test11	structure	line_pipeline	未通过	associate failed: not enough candidate line matches: 0	结构数 81 / 78	0	0	0	0.000	0.000	-	-
Test12	structure	line_pipeline	通过	OK	结构数 13 / 30	0	10	8	0.800	0.000	0.934	0.811

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